Thesis

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April 2026

1 Introduction

Question: To what extent does Neural Posterior Estimation improve the calibration and macro economical forecasting performance of a macroeconomic Agent-Based Model relative to traditional estimation methods?

2 Literature review

In the current state of the art macro economic modelling, there are two main approaches. The first one is based on Dynamic Stochastic General Equilibrium (DSGE) models, which are widely used in macroeconomics for policy analysis and forecasting. Popularized by Smets and Wouters (2007), these models are built on microeconomic foundations and incorporate rational expectations. However, they often rely on strong assumptions and may not capture the complexity of real-world economies.

The second approach, analyzed in this paper, is based on Agent-Based Models (ABMs). First used for economic modelling by Orcutt (1957), ABMs have gained popularity after the financial crisis of 2008. Presenting a more flexible framework than DSGE models, ABMs allow for the inclusion of heterogeneous agents, non-linear interactions, and emergent phenomena. Results by Farmer and Foley (2009), suggest that ABMs can provide valuable insights into economic dynamics and policy implications, especially in situations where traditional models struggle to capture the complexity of the economy.

However, large ABMs often have a large number of parameters, which can make calibration and forecasting challenging. Traditional methods for calibrating ABMs include manual tuning, grid search, and optimization algorithms. These methods can be time-consuming and may not always yield accurate results. Recenly, following the work of Grazini (2017) there has been growing interest in using Bayesian methods for calibration. They present three ABC methods for calibration that mostly rely on constructing an aproximation for the likelihood function. But those taks can be computationally expensive. Dryer et. al (2024) propose two methods for posterior estimation that learn to distuigh between simulation results for different input parameters. Moveover, these methods are consucted to learn one posterior density, reducing the computational burden. They do not impose strong assumptions on the output of the ABM, what makes them significanly more flexible.

The foundation for our analysis is the work of Poledna et. al. (2023), who developed an ABM model of the Austrian economy and demonstrated its forecasting performance. Their calibration method relies on a combination of cross sectional and time series data. Where the parameters for the goverment actions are estimated as coefficient of AR(1) processes. Moreover, my main parameter of interest, the fraction of income devoted to consumption (ψ), is estimated from input - output tables provided by Statistik Austria.

Their model was extended by the Wiese et. al. (2024) paper, where they incorporated Bayesian Method developed by Dyer et. al. (2024) to calibrate some parameters. They show that this method can improve the forecasting performance of the model, especially in the short term. Another application of the Dyer et. al. method is presented by Alves et. al. (2026), where they tune a labour market ABM. They choose three fundamental parameters and find that handcrafted summary statistics result in weaker posterior than NN learned ones.

3 Methodology

We will try to answer the question, whether Neural Posterior Estimation improves the calibration and forecasting performance of ABM models. As the underlying simulation engine, we use ABM model developed by Poledna et al. (2023), based on Italian economy in ($T_1 = 2010Q1$) and calibrated using data from 1995Q1 to 2009Q4. The calibration method is described in the original paper.

Since our analysis is based on forecasting performance, there are two steps:

1. Posterior estimation using NPE method on hypothetical trajectories from 2010Q1 to 2016Q4.
2. Evaluation of forecasting performance of the model with parameters obtained from NPE, compared to the original calibration method and real data.

3.1 Posterior Estimation

The simulation includes a variate of parameters, some of them stay constant, some vary over time. In order to answer the research question, we focused on five constant ones, that we think have significant influence over GDP growth. Moreover we, tried to identify those, for which standard calibration (used by Poledna et. al.) might be difficult. Due to lack of sufficient data or the nature of the parameter itself. Our choice of parameters is presented in Table. 1 and explanation together with GDP decomposition is below:

$$Y = \text{Consumtion} + \text{Investment} + \text{Gov. expenditure} + \text{Net Exports} \quad (1)$$

Symbol	Name	Prior range
ψ	fraction of income devoted to consumption	
α^G	Autoregressive coefficient for government consumption	
α^E	Autoregressive coefficient for exports	
β^E	Scalar constant for exports	
ξ^γ	Weight of economic growth	

Table 1: Parameters to estimate, convention the same as in Poledna et. al paper

ψ - It reflects the fraction of households income devoted to consumption. We decided to include it in our analysis, since consumption is one of the major drivers of GDP growth. While at the same time, the fraction can be quite challenging to estimate.

α^G - Second parameter dictates the persistance of GDP. Since Government expenditure is a significant portion of Austria GDP (more than 50% in 2024), this parameter can identify this relation.

We want to find:

$$p(\theta|x_{\mathrm{obs}}) = \frac{p(x_{\mathrm{obs}}|\theta)p(\theta)}{p(x_{\mathrm{obs}})} \tag{2}$$

But, the likelihood function $p(x|\theta)$ is intractable, we use NPE method. With x_{obs} refering to the observed trajectory of GDP growth from 2010Q1 to 2016Q4. The NPE analysis, as described by Dyer et. al. starts by choosing appropriate priors for the choosen parameters. Following the modeling choice of Alves et. al., and lack of strong signals to deviate, we decided to use uniform priors ($p(\theta)$) with bounds presented in table 1.

With the given priors, we sample $N =$ paremeter draws $\{\theta_i\}_{i=1}^N$. For each of those, we run $M = 10$ simulations of the ABM, starting from T_1 to $T_2 = 2016Q4$. Those M simulations are used to obtain the mean trajectory (3) of the model for given parameter draw θ_i, which we denote as x_i.

$$x_i = \frac{1}{M}\sum_{m=1}^{M} x_i^m \tag{3}$$

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Choice of summary statistics	Motivation
Series mean	
Series standard deviation	
AR(1) coefficient	
Lag 4 (yearly) correlation	
Skewness	

Table 2: Summary statistics

We obtain N pairs $\{\theta_i, x_i\}$. Since x_i is a high - dimensional vector (with the dimension being the forecasting horizon for the training period $T_{train} = T_2 - T_1$ in quaters) We use a set of handcrafted statistics ($n = 10$, table. 2) to reduce variance in the posterior. Moreover, since we are interested in forecasting performance, those statistics can offer more insights about underlying simulation, than raw data. Without those, the NPE might learn spurious relations, or not converge quick enough.

$$s(x_i) : \mathbb{R}^T \to \mathbb{R}^n \tag{4}$$

Therefore, the conditional density estimation problem $p(\theta|x)$ transforms into $p(\theta|s(x))$. Which we estimate using extension of normalizing flows (Tabak and Vanden-Eijnden, 2010; Tabak and Turner, 2013; Rezende and Mohamed, 2015) called Masked Autoregressive Flow (Papamakarios and Murray, 2016; Lueckmann et al., 2017; Papamakarios et al., 2019; Greenberg et al., 2019).

3.2 Evaluation of the posteriors

As with other Bayesian Methods, evaluating the sensibility of the posterior is a crucial step in the process. Since we construct a conditional posterior used for forecasting, we use historical data: from 1995Q1 to 2009Q4, to obtain required summary statistics: $s(x_{obs}) = s(x_{1995Q1:2009Q4})$. With those to condition on, we approach the evaluation in two steps, with sampling from the posterior $p(\theta|s(x_{obs}))$. First we check the posterior itself using conditional densities plots and simulation based calibration - SBC (Talts et. al., 2018 and Cook et al., 2006) to asses its validity. Then we take the mean of the posterior to obtain parameter values used to compute final ABM forecast. It is then compared (using RMSFE) to a) real GDP growth data; b) forecast obtained from the original calibration method based on T_2. The RMSFE is computed for the forecasting horizon $T_{\text{forecast}} = 2017Q1 - 2019Q4$.

4 Results

5 Summary

6 References

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